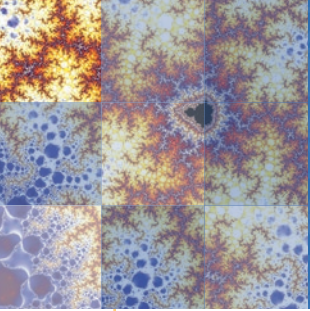




Radical Expressions

An expression that contains a radical sign, $\sqrt{\quad}$, is a **radical expression**.

Examples of radical expressions are $\sqrt{3x}$, $\sqrt[4]{x^2y^3}$, and $\sqrt[5]{32}$.

Definitions

If a and b are real numbers, n is a positive integer, and $a^n = b$, then a is an **n th root** of b .

The n th root of b is written $\sqrt[n]{b}$, where n is the **index** of the radical and b is the **radicand**. The index is always an integer greater than 1, and the radicand can be a real number or an algebraic expression.

► **Think About It** If a and b are real numbers and $a^2 = b$, then a is the square root of b . This also applies to roots other than square roots.

► **Think About It**
NOTATION $\sqrt[n]{b}$ indicates the n th root of b when n is a positive integer.

Finding Real n th Roots

Real n th Roots of b

	n is even	n is odd
$b > 0$	two real roots (one positive and one negative)	one real root
$b < 0$	no real roots	one real root
$b = 0$	one real root, $\sqrt[n]{0} = 0$	one real root, $\sqrt[n]{0} = 0$

EXAMPLE 1

A Find the cube roots of -64 .

SOLUTION

The cube root means that n is 3. Since n is odd and $b < 0$, there is one real root: $(-4)^3 = -64$. The cube root of -64 is -4 .

B Find the fourth roots of -625 .

SOLUTION

Since n is even and $b < 0$, there are no real roots.

C Find the fourth roots of 81 .

SOLUTION

Since n is even and $b > 0$, there are two real roots: $(-3)^4 = 81$ and $3^4 = 81$. The fourth roots of 81 are ± 3 . ■

Simplified Radical Form

You can use properties of radicals to simplify expressions. A square root expression is in simplified radical form if the radicand is not a fraction, there are no radicals in the denominator, and none of the factors is a perfect square other than 1.

Product Property of Radicals

For real numbers $a \geq 0$ and $b \geq 0$,

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}.$$

Quotient Property of Radicals

For real numbers $a \geq 0$ and $b > 0$,

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

► Think About It

NOTATION Assume that the index of a radical expression is 2 whenever no index is written.

EXAMPLE 2

Write the expression in simplified radical form.

A $\sqrt{80}$

SOLUTION

$$\begin{aligned}\sqrt{80} &= \sqrt{16 \cdot 5} && \text{Use a perfect square to factor the radicand.} \\ &= \sqrt{16} \cdot \sqrt{5} && \text{Product Property of Radicals} \\ &= 4\sqrt{5} && \text{Simplify.}\end{aligned}$$

B $\sqrt{\frac{2}{9}}$

SOLUTION

$$\begin{aligned}\sqrt{\frac{2}{9}} &= \frac{\sqrt{2}}{\sqrt{9}} && \text{Quotient Property of Radicals} \\ &= \frac{\sqrt{2}}{3} && \text{Simplify.}\end{aligned}$$

C $\sqrt{15}$

SOLUTION

$\sqrt{15}$ is in simplified radical form because the radicand is not a fraction, there are no radicals in the denominator, and none of the factors (1, 3, 5, or 15) is a perfect square other than 1. ■

Rationalizing a Denominator

When a radical is in the denominator of a fraction, you can simplify it by multiplying both the numerator and denominator by the radical. This process is called rationalizing the denominator.

EXAMPLE 3

Write $\frac{3}{\sqrt{5}}$ in simplified radical form.

SOLUTION

$$\begin{aligned}\frac{3}{\sqrt{5}} &= \frac{3}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} && \text{Multiply the numerator and denominator by } \sqrt{5}. \\ &= \frac{3\sqrt{5}}{\sqrt{5} \cdot 5} && \text{Product Property of Radicals} \\ &= \frac{3\sqrt{5}}{\sqrt{25}} && \text{Simplify.} \\ &= \frac{3\sqrt{5}}{5} && \text{Simplify. } \blacksquare\end{aligned}$$

Simplifying Algebraic Expressions

Here are examples of how you can use the product property of radicals to simplify the square root of a power of a variable.

Let $x \geq 0$.

$$\sqrt{x^2} = x$$

$$\sqrt{x^3} = \sqrt{x^2} \cdot x = \sqrt{x^2} \cdot \sqrt{x} = x\sqrt{x}$$

$$\sqrt{x^4} = \sqrt{(x^2)^2} = x^2$$

$$\sqrt{x^5} = \sqrt{x^4} \cdot x = \sqrt{x^4} \cdot \sqrt{x} = x^2\sqrt{x}$$

EXAMPLE 4

Write the expression in simplified radical form. Assume all variables are positive.

A $\sqrt{27x^2y^3}$

SOLUTION

$$\sqrt{27x^2y^3} = \sqrt{9 \cdot 3 \cdot x^2 \cdot y^2 \cdot y}$$

$$= \sqrt{9} \cdot \sqrt{3} \cdot \sqrt{x^2} \cdot \sqrt{y^2} \cdot \sqrt{y}$$

$$= 3\sqrt{3} \cdot xy\sqrt{y}$$

$$= 3xy\sqrt{3y}$$

Factor the radicand.

Product Property of Radicals

Simplify.

Simplify.

B $\frac{a}{\sqrt{a^3}}$

SOLUTION

$$\frac{a}{\sqrt{a^3}} = \frac{a}{\sqrt{a^2 \cdot a}}$$

Factor the denominator.

$$= \frac{a}{\sqrt{a^2} \cdot \sqrt{a}}$$

Product Property of Radicals

$$= \frac{a}{a\sqrt{a}}$$

Simplify.

$$= \frac{1}{\sqrt{a}} \cdot \frac{\sqrt{a}}{\sqrt{a}}$$

Multiply the numerator and denominator by $\sqrt{a}$.

$$= \frac{\sqrt{a}}{a}$$

Simplify.

C $\sqrt{7hg}$

SOLUTION

$\sqrt{7hg}$ is in simplified radical form because the radicand is not a fraction, there are no radicals in the denominator, and none of the factors is a perfect square other than 1. ■

Multiplying Radical Expressions

To multiply radical expressions, use the product property of radicals.

EXAMPLE 5

Multiply. Assume all variables are positive.

A $\sqrt{8} \cdot \sqrt{32}$

SOLUTION

$$\sqrt{8} \cdot \sqrt{32} = \sqrt{8 \cdot 32}$$

Product Property of Radicals

$$= \sqrt{256}$$

Multiply.

$$= \sqrt{16^2}$$

Identify that $256 = 16^2$.

$$= 16$$

Simplify.

B $\sqrt{2r^3s^9} \cdot 4\sqrt{8r^5s^2}$

SOLUTION

$$\begin{aligned}
 \sqrt{2r^3s^9} \cdot 4\sqrt{8r^5s^2} &= 4\sqrt{2r^3s^9 \cdot 8r^5s^2} && \text{Product Property of Radicals} \\
 &= 4\sqrt{16r^8s^{11}} && \text{Product of Powers Property} \\
 &= 4\sqrt{16r^8s^{10}s} && \text{Product of Powers Property} \\
 &= 4\sqrt{4^2 \cdot (r^4)^2 \cdot (s^5)^2 \cdot s} \\
 &= 4 \cdot 4 \cdot r^4 \cdot s^5 \cdot \sqrt{s} && \text{Simplify.} \\
 &= 16r^4s^5\sqrt{s} && \text{Simplify. } \blacksquare
 \end{aligned}$$

Adding and Subtracting Radical Expressions

Like radicals have the same index and radicand. You can use the distributive property to add or subtract like radicals.

EXAMPLE 6

Add or subtract. Assume all variables are positive.

A $8\sqrt{6} + \sqrt{5} + \sqrt{6}$

SOLUTION

$$\begin{aligned}
 8\sqrt{6} + \sqrt{5} + \sqrt{6} &= 8\sqrt{6} + \sqrt{6} + \sqrt{5} \\
 &= (8 + 1)\sqrt{6} + \sqrt{5} \\
 &= 9\sqrt{6} + \sqrt{5}
 \end{aligned}$$

B $5\sqrt{12} - 3\sqrt{48}$

SOLUTION

$$\begin{aligned}5\sqrt{12} - 3\sqrt{48} &= 5\sqrt{4 \cdot 3} - 3\sqrt{16 \cdot 3} \\&= 5\sqrt{4} \cdot \sqrt{3} - 3\sqrt{16} \cdot \sqrt{3} \\&= 10\sqrt{3} - 12\sqrt{3} \\&= (10 - 12)\sqrt{3} \\&= -2\sqrt{3}\end{aligned}$$

C Simplify.

$$\sqrt{48t^5} - \sqrt{3t^5} - \sqrt{3t}$$

SOLUTION

$$\begin{aligned}\sqrt{48t^5} - \sqrt{3t^5} - \sqrt{3t} &= \sqrt{16t^4} \cdot \sqrt{3t} - \sqrt{t^4} \cdot \sqrt{3t} - \sqrt{3t} \\&= 4t^2\sqrt{3t} - t^2\sqrt{3t} - \sqrt{3t} \\&= (4t^2 - t^2)\sqrt{3t} - \sqrt{3t} \\&= 3t^2\sqrt{3t} - \sqrt{3t} \text{ or } (3t^2 - 1)\sqrt{3t} \quad \blacksquare\end{aligned}$$

► **Think About It** $(3t^2 - 1)\sqrt{3t}$ is an alternate simplified form of $3t^2\sqrt{3t} - \sqrt{3t}$.